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SEAT No. :

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P4

APR - 18/TE/Insem. - 4

T.E. (Civil Engineering)

STRUCTURAL DESIGN - II

(2012 Pattern) (Semester - II)

Time : 1½ Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4 and Q5 or Q6.
- 2) Figures to the right side indicate full marks.
- 3) Use I.S.456-2000 and non programmable calculator is allowed.
- 4) Neat diagrams must be drawn wherever necessary.
- 5) Assume suitable data, if necessary.

Q1) a) Define limit state, and write comment on why limit state is more desirable than working stress method. [5]

b) Explain balanced, under reinforced and over reinforced section w.r.t. LSM. [5]

OR

Q2) a) Derive the design constants used in LSM for M20 and Fe500. [4]

b) Design a RCC beam section using WSM subjected to Bending moment of 85 KN-m using M20 and Fe415. Assume $b = 2d$. [6]

Q3) A RCC beam of size 230 mm $\times$ 525 mm is reinforced with 4 no's of 20 mm dia. Beam having effective span of 5.0 m and clear cover to reinforcement is 30 mm. Calculate safe working UDL including self weight the beam can carry using WSM and LSM, Use M20 and Fe415. [10]

OR

Q4) For an assembly hall 15 m $\times$ 7 m floor beams are spaced at 3.75 m and have a simply supported span of 7.0 m, these beams support a floor slab of 125 mm thick the size of beam is 230 mm $\times$ 525 mm overall, Design the intermediate flanged section for flexure only using LSM. Refer data given below : [10]

P.T.O.

- a) Live load on slab = 4 kN/m^2
- b) Floor finish = 1.5 kN/m^2
- c) Wall on beam = 230 mm thick and 2.6 m height
- d) Effective cover = 50 mm
- e) Material - M20 and Fe500.

Q5) Design a corridor slab over a passage $3 \text{ m} \times 8 \text{ m}$ at the entrance of a public building, the slab is supported by 300 mm wide beams and carries a live load of 4 kN/m^2 and F.F of 1.5 kN/m^2 use M20 and Fe415 design for flexure, shear and development length and show details of reinforcement. **[10]**

OR

Q6) Design the first flight of a dog-legged staircase for the following data : **[10]**

- a) Centre line dimension of staircase room - $2.2 \text{ m} \times 4.75 \text{ m}$.
- b) Floor to floor height = 3.0 m.
- c) Rise = 150 mm Tread = 300 mm.
- d) Width of landing = 1.25 m.
- e) Type of building - Residential with F.F. 1.5 kN/m^2 .
- f) Material - M25 and Fe500.
- g) Draw details of reinforcement.

